The Protective Power of Social Connectedness: Individual vs. Community Influences on Psychological Distress in Middle-Aged Black Adults

This is a multilevel longitudinal and cross-sectional investigation of social connectedness and psychological distress specifically among Black adults, using 10-year follow-up data to distinguish individual versus neighborhood-level effects in an understudied population.

AUTHORS

Yu-Tien Hsu, MD, PhD
Katelyn Kostakis, MPH
Shazel Muhammad
Yusuf Ransome, DrPH, MPH

AFFILIATIONS

Society Connectedness and Health (SOCAH) Lab
Department of Social & Behavioral Sciences
Yale University School of Public Health

01. Introduction

Despite evidence that social connectedness protects mental health, few studies have examined its multilevel impact—at both individual and neighborhood levels—on psychological distress among midlife Black adults, a group disproportionately affected by structural disadvantage and mental health disparities.

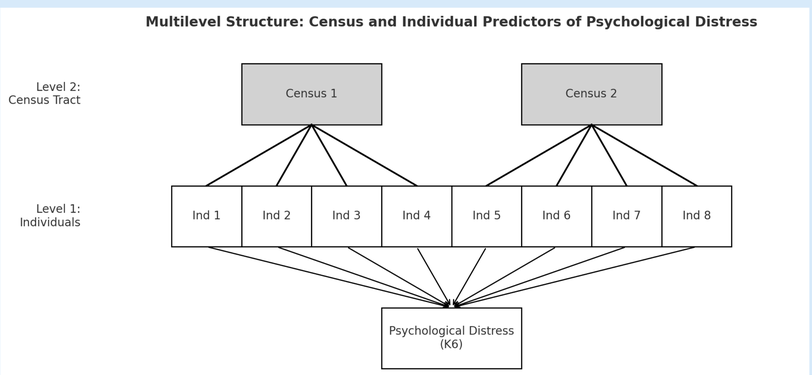
02. Objective

- Investigated how **social connectedness** at both the individual and census tract levels relates to **psychological distress**s among Black midlife adults.
- Used both **longitudinal (MIDUS 2 + MIDUS 3)** and **cross-sectional (MIDUS 3 + Refresher)** data.
- Applied a multilevel modeling approach to examine contextual and individual predictors simultaneously.

03. Main Hypotheses

- Higher social connectedness at both levels would be associated with lower psychological distress.
- Neighborhood-level connectedness was expected to have a stronger effect than individual-level connectedness.

Multilevel Structure: Census and Individual Predictors of Psychological Distress



Acknowledgement

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Declaration of Interests Statement:
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CONTACT US

04. Methods

4.1 Datasets

Longitudinal dataset:
M2: 592 recruited→
M3: 389 followed up i (65.7% retention)

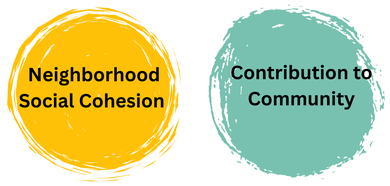
Cross-Sectional dataset:
M3 + MR: 897 in total

4.2 Measures

Individual-Level Social Connectednes



Census-Level Social Connectednes



Covariates

- **Individual-Level:** Age, sex, education, employment status, marital status, income, self-rated physical health, years in neighborhood, and homeownership status
- **Census Tract-Level:**
- Neighborhood composition: Percentage of immigrants and residents aged 65+
- Socioeconomic disadvantage composite: Principal component analysis of poverty rate, education levels, unemployment, rental housing, and median income

4.3 Statistical Analysis

- **Model Selection:** Extensive diagnostics comparing OLS, multilevel, and cluster-robust approaches
- **Primary Method:** Pooled OLS with cluster-robust standard errors (selected due to low ICC <0.05)
- **Longitudinal Models:** Weighted regression and generalized estimating equations (GEE)
- **Attrition Adjustment: Inverse probability weighting (IPW) for 34.3% loss to follow-up**

Model Specifications

- **Model 1:** Individual-level social connectedness only
- **Model 2:** Census tract-level social connectedness only
- **Model 3:** Combined individual + neighborhood levels with full covariates

05. Results

5.1 Study Population

MIDUS 2 Baseline (n=592):
- Mean age: 51.6 years, 62.7% female
- 62.5% high school/some college education

MIDUS 3 Follow-up (n=389)
- Mean age: 61.1 years, 65.3% female

MIDUS Refresher (n=508)
- Younger cohort: Mean age 43.0 years, 56.9% female
- 53.5% never married

Combined Analysis Sample (n=897)
- Black adults, ages 25-74

5.2 Key Findings

Longitudinal Analysis (10-year follow-up)
- **Family support:** Most consistent protective factor across all models (β=-1.76 to -2.37)
- **Friend support:** Significant after IPW adjustment (β=-0.97 to -1.09)
- **Community contribution:** Unexpected positive association with distress (β=1.32-1.35)

Cross-sectional Analysis
- **Neighborhood social cohesion:** Strong protective effect (β=-1.44 to -2.38)
- **Family support:** Consistent protective effect (β=-1.83 to -2.62)
- **Friend support:** Significant protective effect (β=-0.95 to -1.16)

***Highlight: Individual-level social connectedness** showed stronger associations than neighborhood-level measures

06. Conclusions

Individual > Neighborhood Effects
- Family and friend support showed strongest protective effects against psychological distress.
- Prioritize individual social network interventions

Community Engagement: Long-Term Benefits
- Community contribution showed protective effects only over time
- → Support sustained community involvement initiatives for cumulative mental health benefits